Keeping Alternatives Alive: A Comparison-Preserving Scaffold for Root Cause Analysis Under Closure Pressure

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Abstract

Root cause analysis (RCA) in pharmaceutical manufacturing is designed to identify systemic causes — yet collaborative investigations routinely converge on individual blame before alternatives have been meaningfully compared. This pattern reflects a property of the investigative system rather than an individual reasoning failure: closure pressure and documentation requirements collapse the comparison space early, before systemic contributors can be meaningfully examined.

Drawing on Premature Accountability Convergence (PAC) — a framework describing how closure pressure accelerates how responsibility settles during investigation — I ask the inverse design question: can AI expand interpretive space rather than compress it?

I report a two-study empirical program with N=20 GMP-experienced participants under time-constrained investigative conditions. Study 1 shows that convergence is structural and time-sensitive: initial explanations were uniformly single-cause and person-centered, but broadened with additional reflection time. Study 2 introduces an AI-supported reasoning scaffold that functions as a comparison enforcer — not by suggesting causes, but by detecting premature convergence signals and triggering reopening prompts. The scaffold increased causal breadth, sustained alternative exploration, and reduced person-centered attribution. The most significant adoption barrier was organizational, raising tensions between transparency and existing accountability practices.

I introduce comparison preservation as a design principle for AI-supported reasoning in accountability-critical environments, and argue that accountability in collaborative investigative work is structured before attribution — and that design can reopen the space where that structuring occurs.

Keywords

root cause analysis, accountability, closure pressure, human-AI interaction, comparison preservation, regulated environments

1 Introduction

Root cause analysis (RCA) in pharmaceutical manufacturing is a formal quality requirement embedded in collaborative investigative work: every deviation from an approved process must be investigated, its cause documented, and corrective actions assigned before the record can be closed. In principle, this investigation should compare multiple plausible causes before converging on an explanation. In practice, investigations unfold under closure pressure — temporal deadlines, documentation requirements, and institutional expectations that define when an investigation must be considered complete. Under these conditions, the first viable explanation tends to become the documented one, not because alternatives have been eliminated, but because the workflow does not keep them comparable long enough to be evaluated.

Prior work on investigation failure has largely framed this as a cognitive problem: investigators rely on recognition-primed decision making [4], anchor on initial hypotheses, or fail to generate sufficient alternatives. I argue this framing misidentifies the locus of failure. Premature convergence in regulated investigations is not primarily a property of how individuals reason — it is a property of how investigative workflows organize, interrupt, and foreclose comparison under institutional pressure. The comparison space collapses before alternatives can be meaningfully compared within the workflow, not because investigators lack skill, but because the system does not structurally require comparison to be sustained.

This paper introduces comparison preservation as a design principle for AI-supported reasoning in accountability-critical environments. Drawing on Premature Accountability Convergence (PAC) — a framework describing how closure pressure accelerates how responsibility settles during investigation, shaping how explanations are generated, shared, and stabilized across actors — I ask the inverse design question: rather than accelerating convergence, can AI be designed to expand and maintain the interpretive space within which comparison occurs? I report a two-study empirical program that first establishes the structural and time-sensitive nature of convergence collapse, then introduces and evaluates an AI-supported scaffold designed to enforce comparison before closure.

This paper makes three contributions. First, it reframes premature convergence as a failure of investigative system design rather than individual cognition, grounding this claim in empirical observation of how reasoning develops under closure pressure. Second, it introduces the comparison enforcer as a design role for AI in collaborative investigative settings — distinct from suggestion engines or automated classifiers. Third, it identifies organizational transparency as an adoption barrier specific to regulated environments, where making reasoning visible surfaces accountability conditions that existing practice leaves invisible.

2 Related Work

2.1 Causal Explanation as Organizational Outcome

Vaughan's account of the Challenger launch decision establishes that causal explanations in high-stakes organizations reflect the conditions of their production — not only the evidence available, but the workflow structures, institutional pressures, and normalization processes under which deliberation unfolds [8]. Klein's account of recognition-primed decision making shows that experts under

time pressure do not compare alternatives but recognize and act on the first viable option [4]. Together, these accounts suggest that premature convergence in investigation is structurally produced — yet neither addresses how workflow design might intervene to keep comparison open — a gap this paper addresses by introducing comparison preservation as a design intervention.

2.2 Invisible Work and Coordination in Regulated Systems

Star and Strauss show that coordination work performed in the gaps of formal systems becomes invisible — not captured by indicators, not attributed to the structural conditions that produce it [6]. In regulated investigation workflows, the comparative work of holding multiple causal accounts in parallel is precisely this kind of invisible labor: unrewarded by closure expectations and absent from final investigation records. Suchman further establishes that action is shaped by the immediate conditions of its execution — what is visible and interpretable at the moment of decision [7]. Prior CSCW research has examined invisible work and awareness in collaborative settings [2], but has not addressed how comparison collapse in accountability-critical workflows can be made visible and structurally required — a gap this paper addresses by making comparison work explicitly visible and enforceable.

2.3 Human-AI Interaction and Judgment-Preserving Design

Amershi et al. formalize eighteen guidelines for human-AI interaction, emphasizing that AI systems should support human control and avoid pre-empting judgment [1]. In regulated environments, this principle takes on particular force: quality decisions must remain with qualified human reviewers, and systems that automate conclusions introduce compliance risk. Prior work has examined AI as a source of suggestions or classifications — a model that accelerates convergence by anchoring reasoning around computationally foregrounded interpretations. This paper reframes the role of AI from suggesting or prioritizing explanations to enforcing the sustained comparability of alternatives under closure pressure.

2.4 Prototype-Based Evaluation

Nielsen demonstrates that think-aloud evaluation with small participant samples reliably surfaces usability and reasoning support properties [5]. Hutchinson et al. establish the logic of the technology probe — a prototype used simultaneously as a design intervention and an analytic instrument, making visible the conditions it is designed to address [3]. This study applies both approaches to regulated investigation reasoning, allowing the analysis to not only evaluate usability but to surface otherwise invisible dynamics of convergence under real workflow constraints.

3 Method

3.1 Study Context

Both studies were conducted in a pharmaceutical manufacturing quality system operating under GMP regulations. GMP-regulated environments impose significant access constraints on participant recruitment: qualified investigators with direct deviation investigation experience represent a specialized population within a compliance-bound facility. N=20 participants were recruited from a single pharmaceutical quality team with direct experience in deviation investigation, root cause analysis, and CAPA documentation. While drawn from a single team, participants represented varied levels of experience and roles within the investigative workflow, allowing observation of convergence patterns across heterogeneous expertise. This sample size is consistent with established think-aloud methodology [5] and reflects the structural constraints of conducting research within a regulated organizational setting. The goal was not statistical generalization, but to surface structural patterns under controlled investigative conditions.

Both studies used the same standardized deviation scenario — an out-of-specification analytical result with sufficient contextual detail to support multiple plausible causal interpretations, including person-centered, procedural, equipment-related, and organizational accounts.

3.2 Study 1: Timed Reflection Protocol

Participants produced causal explanations suitable for formal documentation at three timepoints: immediately after reading the scenario (T0), after five minutes of unstructured reflection (T+5), and after ten minutes (T+10). Responses were analyzed for causal breadth, causal type (person-centered, procedural, systemic, or mixed), and evidence of comparison across alternatives. The protocol was designed to isolate whether convergence reflects fixed reasoning limitations or constraints imposed by investigative conditions.

3.3 Study 2: Comparison-Preserving Scaffold

Participants completed the same scenario using an AI-supported reasoning scaffold implemented as a Streamlit web application. The scaffold organized investigation into five stages — case intake, hypothesis expansion, PAC detection, structured comparison, and pre-closure verification — enforcing explicit comparison before closure. Participants interacted with the scaffold through guided prompts and structured input fields, ensuring that alternative explanations were explicitly articulated before comparison. The AI component functioned exclusively as a comparison enforcer: detecting convergence signals and triggering reopening prompts, without suggesting causes or generating conclusions. Post-task measures were derived from structured outputs generated through the scaffold, including causal breadth (number of distinct causal categories), evidence of explicit comparison between alternatives, and interaction traces reflecting when and how participants revisited or expanded initial explanations. Post-task discussion probed organizational barriers, particularly tensions between reasoning transparency and existing accountability practices.

4 Results

4.1 Study 1: Convergence as Structural and Time-Sensitive

At T0, all participants produced single-cause explanations, most of which were person-centered. With additional reflection time (T+5,

T+10), explanations expanded to include procedural, equipment-related, and organizational factors. Participants increasingly introduced alternative accounts over time, indicating that causal breadth was not fixed but developed when comparison was sustained. This pattern suggests that early convergence reflects constrained investigative conditions rather than limited reasoning capacity.

4.2 Study 2: Effects of the Comparison-Preserving Scaffold

Under the scaffold, participants produced more multi-category explanations and engaged in explicit comparison between alternatives. The scaffold prompted participants to revisit initial explanations and articulate competing accounts before closure. Compared to Study 1, person-centered explanations were less dominant, and causal accounts more frequently incorporated procedural and systemic factors. These results indicate that enforcing comparison can expand causal reasoning and delay premature convergence.

4.3 Adoption Barrier: Organizational Transparency Tension

Despite positive interaction outcomes, participants raised concerns about organizational adoption. In particular, making reasoning processes visible was perceived as introducing accountability risk, as intermediate or alternative explanations could be subject to scrutiny. This highlights a tension specific to regulated environments: systems that support better reasoning may conflict with existing accountability practices that favor stable, defensible conclusions.

5 Discussion

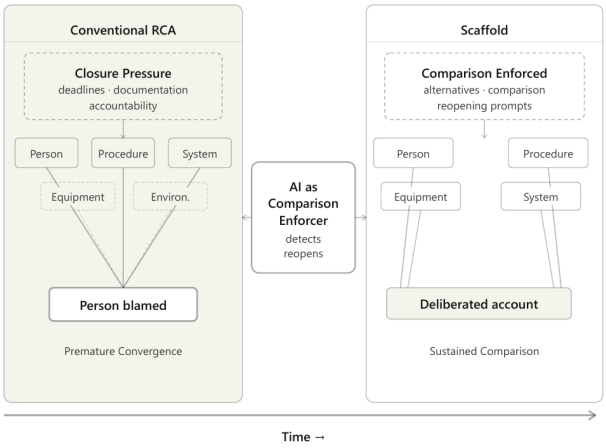

Figure 1: Premature Accountability Convergence (left) versus a comparison-preserving scaffold (right). Under closure pressure, conventional RCA converges early, collapsing the comparison space before alternatives can be meaningfully evaluated. The scaffold preserves this space by sustaining comparison across multiple alternatives prior to closure.

5.1 Convergence as a System Property

The findings show that premature convergence is a property of investigative systems rather than individual reasoning. Investigators are capable of broader reasoning, but existing workflows do not sustain the conditions required for comparison. Closure pressure and documentation requirements produce convergence before alternatives can be meaningfully evaluated, shifting the design target from the individual to the structure of the investigative workflow.

5.2 PAC as an Empirical Dynamic

These findings empirically demonstrate Premature Accountability Convergence (PAC) as a temporal dynamic. Under closure pressure, convergence occurs early — but can be delayed when comparison is structurally enforced. Study 1 shows that causal breadth develops over time, while Study 2 demonstrates that design interventions can expand the window in which alternative explanations remain viable.

5.3 Comparison Preservation and Organizational Tension

The comparison enforcer reframes the role of AI in accountability-critical environments: not to accelerate conclusions, but to sustain comparability before attribution. However, the identified adoption barrier reveals a deeper tension — making reasoning visible may conflict with existing accountability practices that favor stable, defensible conclusions. This suggests that improving investigative reasoning may require redesigning not only tools, but the accountability structures that govern how reasoning is evaluated.

6 Conclusion

Premature convergence in regulated investigation is not a reasoning failure, but a property of investigative systems under closure pressure. This work shows that comparison collapse is time-sensitive — and therefore designable. An AI-supported scaffold that enforces explicit comparison before closure can expand causal breadth and delay premature attribution. The challenge is not to improve AI accuracy, but to preserve the conditions under which human judgment remains comparative. This requires engaging with the organizational structures that currently make premature convergence rational.

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